

BA

NE 763 001 – FALL 2026 – BEELER

Exam #1 (Paper)

Student's Name: **Banerjee, Souri**

Date of Exam: Friday, September 4, 2026 - Friday, September 18, 2026

Time Limit: ~~120 minutes~~ 180 minutes (1.5x time)

Appointment Start Time: 1:00pm, Wednesday, Sep 9, 2026

Appointment End Time: 4:00pm, Wednesday, Sep 9, 2026

Allowed Materials: Scientific Calculators

Student Accommodations: Extended Time (1.5T)

**Academic
Testing Services**

Student – NC State University Pack Pledge

I, **SOURI BANERJEE** have neither given nor received unauthorized aid on this exam or assignment. I have read the instructions and acknowledge that this is the correct exam.

STUDENT'S PRINTED NAME

Sour Banerjee
STUDENT SIGNATURE

9/9/26

DATE

Exam must be turned in by: **4:17**

EXAM END TIME

SB

STUDENT'S
INITIAL
AGREEMENT

**NOTE: Failure to turn in exam
on time may result in penalties
at the instructor's discretion.**

SEP 9 4:17pm
MAC

KC MI
kernel

mechanical interaction

NE 763 Exam 1

Point values indicated expected depth of response.

1. What are the individual layers in a TRISO particle? What purpose does each layer have? (10 pts)
2. What are the trade-offs between carbide and oxide fuel kernels? (12 pts)
3. Discuss fission product chemistry in the fuel kernel. (8 pts)
 UO_2 generates CO/CO_2 gas. UC does not retain gas product FP (metals, insoluble/soluble oxides, volatile noble gases)
4. How does the nature of irradiation damage accumulation change with irradiation temperature in SiC? How does this impact swelling? (18 pts)
? < 473
 $473 - 1273$
5. How does thermal conductivity change in SiC with irradiation? What are the primary phonon scatterers? (10 pts)
? SiC w/ irradiation, more ductile, thermal conductivity decreases
6. How does graphite dimensionally change under irradiation? Why does this behavior happen? (18 pts)
more negative? $\rightarrow$ C - expansion a - contraction $\rightarrow$ as lattice closes
C-expansion happens faster $\rightarrow$ under irradiation?
7. List and explain three types of failure mechanisms for TRISO particles. (9 pts)
but more waste $\rightarrow$ overpressure, KC MI, IP/C-SiC debonding
8. Provide one example of an advanced TRISO concept and explain why it is of interest. (5 pts)
 $\rightarrow$ ZrC instead of SiC
ZrC/SiC
9. What are the concerns for intermediate heat exchanger materials in a He environment? Discuss corrosion of these materials. (10 pts)
Alloy 617 - good for high temps, Nickel based, chromia layer protecting from environmental damage/degradation
corrosion

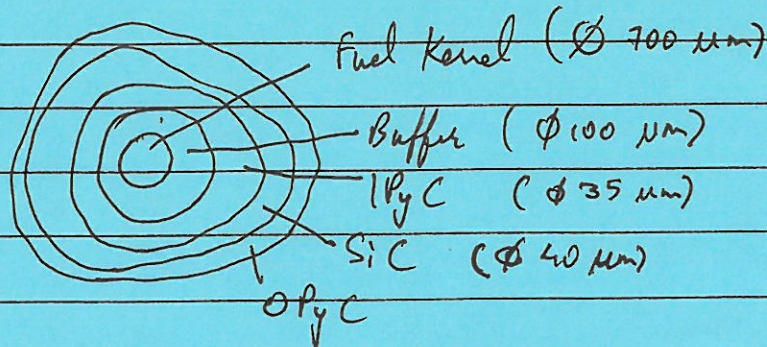
< 473 - amorphize

$473 - 1273$ -

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- 1) Fuel Kernel - Has fissile/fertile material - This is where the fission process takes place
 Buffer - has voids to allow fission gases prevent fission recoil. Has void to help w/ swelling.
 IPyC (inner Pyrolytic carbon) - prevents fission products (FP) from getting to SiC
 SiC - Primary pressure vessel and FP (fission product) barrier
 OPyC (outer Pyrolytic carbon) - to make TRISO manufacturing process "easier". Prevents SiC from getting damaged during galvanostatic process.



- 2) $^{235}\text{UO}_2$ has free oxygen than makes CO gas and can cause excessive pressure. But it retains fission products (FP).
 • UC does not generate CO and other gases preventing excessive pressure, however, fission products are not retained.

In the US, TRISO particles are made of UCO , 80% UO_2 / 20% UC , which is an improvement ^{upon} UO_2 and UC , it doesn't generate CO gas and it retains fission products.

- 4) Swelling in SiC is dependent on temperature / hence As irradiation temp Below 473K the SiC is amorphized. From 473 - 1273K the SiC becomes more elastic. As the temps go up the irradiation ~~accumulation~~ damage accumulation

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(fluence) goes ^{down} ~~down~~ keeping the swelling under control. Too much radi. Increased irradiation damage accumulation can cause swelling to increase.

3) VO_2 generates CO/CO_2 gas. UC does not generate CO gas but does not retain fission products (FPs). FPs such as insoluble and soluble oxides, metals, volatiles, noble gases.

Eg of metal: Ag

Eg of volatile: Cs

Eg of noble gas: Kr

Eg of insoluble oxide: Y ; Eg of soluble oxide: Br, Rb

4) He by itself is a noble gas but it is used as a coolant in ~~HTGR~~ ^{sometimes (400-1000°C)} (High Temp gas Rx) operating at at least 750°C . The concern is finding a material for the IHX (intermediate heat exchanger) that can handle high temps, handle environmental degradation for the operating life of Rx (60 yrs).

A material that is well suited for this is Alloy 617 - its good for high temps, it's nickel based ~~and~~ and it depends on a chromia layer that will protect the ~~major~~ rest of the material from environmental degradation.

We also need a material w/ high thermal conductivity which is a concern Alloy 617 solves

7) Overpressure can lead to IPyC cracking, leading to SiC cracking, when the OPyC will then also crack. IPyC - SiC debonding can cause the SiC to stop being in compression causing the SiC to crack easier. KCM1 (Kernel chemical mechanical interaction), the fuel kernel is in a graphite matrix and

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the kernel may separate and contact the buffer. This is bad because now the ~~fission~~ product will be kernel will be fission in the buffer and the fission recoil and ^{fission} products will be considerably closer to the IP and SiC causing ~~an increased~~ stresses to increased stresses and potential corrosion.

8) ZrC instead of SiC or ZrC/SiC, any advanced TRISO concept is to mitigate ~~current~~ ^{current} concerns there might be w/ the current vision of TRISO. Being able to handle higher temps and higher irradiation damage accumulation, easier manufacturability etc. People are also looking at making TRISO more economically viable and reusable and potentially recovering used fuel for ~~beneficial purposes~~ "beneficial / good-faith purposes". There was some

5) As irradiation increases, ~~rather~~ SiC becomes more ductile, and thermal conductivity ~~decreases~~ ^{increases}. Primary phonon scatterers include alpha and ~~beta~~ ^{beta} particles. Excess irradiation may cause voids and swelling.

6) With graphite under irradiation, we ~~may~~ ^{may} see point defects/interstitials and black spots. ~~The~~ Dimensionally, graphite experiences expansion ~~in~~ ⁱⁿ "c" and contraction in "a". Additionally, as lattice ~~close~~ ^{close} the "c" expansion happens ~~not~~ ^{not} faster. This is called shrinkage and turnaround, this ~~probably~~ happens due to defects such as point defects (black spots), ~~inter~~ interstitials and voids.

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Q) Continued after ~~Q3~~^{Q4} from pg 2, apart from environmental degradation the corrosion the IHX could be subject to depends on the ^{other} coolant in the IHX. Molten salts are quite corrosive as well as sodium. Pb-Bi eutectic coolants are less corrosive. If ~~the~~ water is used, then we'd need to worry about ~~corrosion~~ and crud from oxides formed over time from H_2O . To prevent impurities from getting into He and then damaging the IHX over time, great care must be used to filters and treatment must be used to ensure both the Helium and other coolant is "pure" to an acceptable industry standard.

Q) Continued from Q4 earlier, the irradiation damage accumulated ~~over~~ decreases w/ the increase of irradiation temperature. This is a key factor in keeping swelling in check. ~~the~~ SiC would swell less w/ increased irradiation temp over time due to it accumulating less irradiation damage.

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